Clearance & Elimination Practice 1

Module 4 - Drug Elimination and Clearance · Daily practice sheet · No date

Name _____ Date _____

(1st problem, unnumbered)

Estimate the CrCl of a 36-year-old male who weighs 160 lb, S_{Cr} =0.8 mg/dL, and 68” tall.

a The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. Calculate the half-life of the agent.

b The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. What is the amount of the agent in the body when the concentration of drug in the plasma is 60 mg/L?

c The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. What is the rate of elimination (mg/hr) when the concentration of drug in the plasma is 50 mg/L?

d The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. What is the rate of elimination (mg/hr) when the concentration of drug in the plasma is 25 mg/L?

e The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. What is the expected plasma concentration 12 hours after a 700-mg intravenous bolus dose of the agent?

f The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. Urine was collected for 96 hours and 25 mg of unchanged the agent was recovered from the original 700-mg dose. What is the fraction of drug excreted in the urine?

g The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. Urine was collected for 96 hours and 25 mg of unchanged the agent was recovered from the original 700-mg dose. What is the excretion rate constant of the agent?

h The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. Urine was collected for 96 hours and 25 mg of unchanged the agent was recovered from the original 700-mg dose. What is the renal clearance of the agent?

i The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. Urine was collected for 96 hours and 25 mg of unchanged the agent was recovered from the original 700-mg dose. What is the metabolic (hepatic) clearance of the agent?

j The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. Urine was collected for 96 hours and 25 mg of unchanged the agent was recovered from the original 700-mg dose. What AUC would you expect from the 700-mg dose?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

(1st problem, unnumbered) Her key

Answer: 123.5 mL/min

WHAT THE STEM GIVES

age	36-year-old	140 − 36 = 104 on top
sex	male	IBW = 50 + 2.3 × inches over 5 ft; no 0.85 factor
weight	160 lb	*not needed: her keys use ideal body weight, not actual weight*
S_{Cr}	0.8 mg/dL	the denominator, 72 × 0.8 = 57.6
height	68”	68 − 60 = 8 inches over 5 ft, so IBW = 68.4 kg

THE WORKING, ONE LINE AT A TIME

UNIT

$$68 \text{ in} - 60 \text{ in} = 8 \text{ inches over 5 ft}$$

The ideal body weight equation counts only the inches above five feet, so 60 inches are subtracted from the height.

ALGEBRA

$$\text{IBW} = 50 + 2.3(8) = 68.4 \text{ kg}$$

The male form starts at 50 kg. The 160 lb total body weight is not used, because her keys use ideal body weight.

SETUP

$$\text{CrCl} = \frac{(140 - 36)(68.4)}{(72)(0.8)}$$

This is the Cockcroft-Gault equation; the patient is male, so the 0.85 factor for a female patient is not applied.

ROUND

$$\text{CrCl} = 123.5 \text{ mL/min}$$

(104)(68.4) = 7113.6 on top and (72)(0.8) = 57.6 below; $\frac{7113.6}{57.6}$ = 123.5 mL/min. CrCl, the creatinine clearance, is reported in mL/min, the unit of the glomerular filtration rate it estimates, to one decimal place as her key does.

> **DOES THE NUMBER MAKE SENSE?**
> A 36-year-old man with a normal S_{Cr} of 0.8 mg/dL has a normal GFR, about 120 mL/min, so a CrCl near 120 mL/min fits; a value far below 100 or above 140 means a slipped number.

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a Her key

Answer: 12.5 hr

WHAT THE STEM GIVES

Cl	0.5 L/hr	the denominator of $t½ = \frac{0.693\ V_D}{Cl}$
V_D	9 L	the numerator, 0.693 × 9 = 6.237 L

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½ = \frac{0.693\ V_D}{Cl}$$

$Cl = k \times V_D$ and $k = \frac{0.693}{t½}$ combined give $t½ = 0.693\ \frac{V_D}{Cl}$, so the half-life comes straight from volume and clearance.

ALGEBRA

$$t½ = \frac{0.693(9\ L)}{0.5\ L/hr} = 12.5\ hr$$

0.693 × 9 L = 6.237 L on top; dividing by Cl = 0.5 L/hr, the litres on top cancel the litres in L/hr and the reciprocal hours invert to hours: $\frac{6.237\ L}{0.5\ L/hr} = 12.47$ hr, reported as 12.5 hr, the half-life t½ of the agent.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance of 0.5 L/hr empties the 9 L volume once in 18 hours; a half-life is 0.693 of that turnover time, so about 12.5 hr. A half-life of a few minutes or a few days means a slipped unit.

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b

Her key

Answer: 540 mg

WHAT THE STEM GIVES

Cl	0.5 L/hr	*not needed: amount is concentration times volume*
V_D	9 L	multiplies the concentration: $D_B = C_p \times V_D$
C_p	60 mg/L	the concentration in each of the 9 litres

THE WORKING, ONE LINE AT A TIME

SETUP

$$D_B = C_p \times V_D$$

Concentration is amount divided by volume, $C_p = \frac{D_B}{V_D}$, so multiplying the concentration by the volume gives the amount in the body.

ALGEBRA

$$D_B = (60\ mg/L)(9\ L) = 540\ mg$$

C_p, the plasma concentration, is 60 mg/L and V_D, the volume of distribution, is 9 L; (60 mg/L)(9 L) = 540 mg. The litres of V_D cancel the litres under mg, so an amount in milligrams is left: D_B, the drug in the body when the plasma reads 60 mg/L.

> **DOES THE NUMBER MAKE SENSE?**

> Each of the 9 litres holds 60 mg, so 9 × 60 = 540 mg; the amount must be larger than 60 mg because the volume is more than 1 L.

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c Her key

Answer: 25 mg/hr

WHAT THE STEM GIVES

Cl	0.5 L/hr	multiplies the concentration: rate = Cl × C_p
V_D	9 L	*not needed: the rate uses clearance, not volume*
C_p	50 mg/L	the concentration being cleared

THE WORKING, ONE LINE AT A TIME

SETUP

Rate of elimination = (Cl)(C_p)

Clearance Cl is the volume of plasma cleared of drug per hour, 0.5 L/hr, and each litre cleared carries C_p = 50 mg of drug, so the amount removed per hour is Cl × C_p. The rate of elimination is therefore clearance times the plasma concentration, in mg/hr.

ALGEBRA

Rate = (50 mg/L)(0.5 L/hr) = 25 mg/hr

(50 mg/L)(0.5 L/hr): the litres under mg cancel the litres in L/hr, leaving mg/hr, so 50 × 0.5 = 25 mg of drug leaves the body each hour while the concentration is 50 mg/L. The rate falls as C_p falls; Cl itself does not change.

> **DOES THE NUMBER MAKE SENSE?**
> Half a litre is cleared each hour and each litre holds 50 mg, so 25 mg leaves per hour; the rate is smaller than 50 because Cl is below 1 L/hr.

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d Her key

Answer: 12.5 mg/hr

WHAT THE STEM GIVES

Cl	0.5 L/hr	multiplies the concentration: rate = Cl × C_p
V_D	9 L	*not needed: the rate uses clearance, not volume*
C_p	25 mg/L	the concentration being cleared, half that of part c

THE WORKING, ONE LINE AT A TIME

SETUP

Rate of elimination = (Cl)(C_p)

Clearance is constant at 0.5 L/hr, so the rate of elimination is again clearance times the plasma concentration.

ALGEBRA

Rate = (25 mg/L)(0.5 L/hr) = 12.5 mg/hr

Clearance is the same as in part c, so half the concentration gives half the rate: elimination rate is proportional to concentration.

> **DOES THE NUMBER MAKE SENSE?**
> Half a litre cleared each hour, 25 mg in each litre, so 12.5 mg/hr; half the concentration of part c (50 mg/L) gives half the rate of 25 mg/hr.

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e Her key

Answer: 40 mg/L

WHAT THE STEM GIVES

Cl	0.5 L/hr	with V_D gives $k = \frac{0.5}{9} = 0.0556\ hr^{-1}$
V_D	9 L	$C_0 = \frac{700}{9} = 77.78$ mg/L, and the denominator of k
t	12 hours	the exponent $kt = 0.0556 \times 12$
D_0	700-mg	the numerator of $C_0 = \frac{D_0}{V_D}$
route	intravenous bolus	F = 1 and the whole dose is C_0 at time zero

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{Cl}{V_D} = \frac{0.5\ L/hr}{9\ L} = 0.0556\ hr^{-1}$$

$Cl = k \times V_D$ rearranged gives $k = \frac{Cl}{V_D}$; litres cancel and reciprocal hours are left, the unit a rate constant needs.

SETUP

$$C = \frac{D_0}{V_D}\, e^{-kt}$$

After an IV bolus the concentration starts at $\frac{D_0}{V_D}$ and falls exponentially with the first-order rate constant.

ALGEBRA

$$C12 = \frac{700\ mg}{9\ L}\, e^{-(0.0556)(12)} = (77.78)(0.513)$$

$C_0 = \frac{700\ mg}{9\ L} = 77.78$ mg/L is the concentration at time zero, D_0 over V_D. The exponent $kt = (0.0556\ hr^{-1})(12\ hr) = 0.667$ is a pure number because the hours cancel, and $e^{-0.667} = 0.513$ is the fraction of C_0 still present at 12 hours.

ROUND

C12 = 40 mg/L

(77.78 mg/L)(0.513) = 39.9 mg/L, reported as 40 mg/L. The half-life is $t_{½} = \frac{0.693}{0.0556\ hr^{-1}} =$ 12.5 hr, so 12 hours is just under one half-life and just over half of the 77.78 mg/L starting concentration remains.

> **DOES THE NUMBER MAKE SENSE?**
> C_0 is $\frac{700}{9}$ = 77.78 mg/L, and e^{-kt} = 0.513 is a little more than half, since 12 hours at k = 0.0556 hr^{-1} is just under one half-life. The answer must lie below 77.78 mg/L.

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f

Her key

Answer: 0.036 (fraction)

WHAT THE STEM GIVES

Cl	0.5 L/hr	*not needed: f_e is a ratio of amounts*
V_D	9 L	*not needed*
collection	96 hours	*not needed for the number; long enough to count the recovery as $D_u\infty$*
$D_u\infty$	25 mg	the numerator, unchanged drug recovered
D_0	700-mg	the denominator, with F = 1 for the IV dose

THE WORKING, ONE LINE AT A TIME

SETUP

$$f_e = \frac{D_u\infty}{D_0}$$

f_e, the fraction excreted unchanged, is $D_u\infty$ over F D_0: the unchanged drug recovered in urine, $D_u\infty$, over the amount that reached the circulation, F × D_0. The 700 mg dose was intravenous, so F, the bioavailable fraction, is 1 and the whole dose is the denominator.

ALGEBRA

$$f_e = \frac{25\ mg}{700\ mg} = 0.036$$

25 mg recovered over the 700 mg dose: the milligrams cancel, so f_e is a pure number, $\frac{25}{700}$ = 0.0357, rounded to 0.036. Only 3.6% of the dose leaves in urine unchanged; the remaining 96.4% is removed by another route, metabolism.

> **DOES THE NUMBER MAKE SENSE?**
> 25 mg of the 700 mg dose: 25 is one twenty-eighth of 700, so f_e is about 0.036. A fraction lies between 0 and 1; a value above 1 means the dose and the amount recovered were swapped.

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g

Her key

Answer: 0.002 hr^{-1}

WHAT THE STEM GIVES

Cl	0.5 L/hr	with V_D gives $k = \frac{0.5}{9} = 0.0556\ hr^{-1}$
V_D	9 L	the denominator of $k = \frac{Cl}{V_D}$
collection	96 hours	*not needed*
$D_u\infty$	25 mg	with the dose gives $f_e = \frac{25}{700} = 0.036$
D_0	700-mg	the denominator of f_e

THE WORKING, ONE LINE AT A TIME

SETUP

$k_e = f_e \times k$

$f_e = \frac{k_e}{k}$ rearranged gives $k_e = f_e \times k$: the excretion rate constant is the renal share of the overall rate constant.

ALGEBRA

$k_e = (0.036)(0.0556\ hr^{-1}) = 0.002\ hr^{-1}$

k, the overall elimination rate constant, is $\frac{Cl}{V_D} = \frac{0.5\ L/hr}{9\ L} = 0.0556\ hr^{-1}$, the value found in part e; $f_e = 0.036$ is the fraction excreted unchanged and has no units, so $k_e = (0.036)(0.0556\ hr^{-1}) = 0.0020\ hr^{-1}$ keeps the reciprocal hours.

> **DOES THE NUMBER MAKE SENSE?**
> k_e is the renal share of k: $f_e = 0.036$ of $k = 0.0556\ hr^{-1}$, so about $0.002\ hr^{-1}$. k_e must be smaller than k, $0.0556\ hr^{-1}$, because only part of elimination is renal.

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h

Her key

Answer: 0.018 L/hr

WHAT THE STEM GIVES

Cl_T	0.5 L/hr	multiplied by f_e: $Cl_R = f_e \times Cl_T$
V_D	9 L	*not needed: clearance is given, so no k enters*
collection	96 hours	*not needed*
$D_u\infty$	25 mg	with the dose gives $f_e = \frac{25}{700} = 0.036$
D_0	700-mg	the denominator of f_e

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_R = f_e \times Cl_T$

Total clearance Cl_T, 0.5 L/hr, is the volume of plasma cleared of drug per hour by every route together. The kidney accounts for the fraction $f_e = 0.036$ of the drug eliminated, so the renal clearance Cl_R is that fraction of Cl_T: $Cl_R = f_e \times Cl_T$.

ALGEBRA

$Cl_R = (0.036)(0.5\ L/hr) = 0.018\ L/hr$

(0.036)(0.5 L/hr) = 0.018 L/hr. f_e is a pure number, so multiplying by it changes the size, not the unit, and Cl_R stays in L/hr: of every 0.5 L of plasma cleared each hour, the kidney accounts for the equivalent of 0.018 L.

DOES THE NUMBER MAKE SENSE?

The kidney handles $f_e = 0.036$ of the 0.5 L/hr total, so Cl_R is about 0.018 L/hr, far below the total. Cl_R cannot exceed Cl_T, 0.5 L/hr.

Her key prints the unit as "L/hr^{-1}"; the unit is L/hr.

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i

Her key

Answer: 0.482 L/hr

WHAT THE STEM GIVES

Cl_T	0.5 L/hr	the total: $Cl_H = Cl_T - Cl_R$
V_D	9 L	*not needed*
collection	96 hours	*not needed*
$D_u\infty$	25 mg	with the dose gives $f_e = 0.036$, so $Cl_R = 0.036 \times 0.5 = 0.018$ L/hr
D_0	700-mg	the denominator of f_e

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_H = Cl_T - Cl_R$

Total clearance is renal plus hepatic, so subtracting the renal part from the total leaves the hepatic (metabolic) part.

ALGEBRA

$Cl_H = (0.5 - 0.018)\ L/hr = 0.482\ L/hr$

0.5 L/hr − 0.018 L/hr = 0.482 L/hr; both terms are in L/hr, so the difference is too. The hepatic (metabolic) clearance Cl_H is 0.482 of the 0.5 L/hr total, the fraction $1 - f_e = 0.964$, so metabolism removes almost all of this drug.

DOES THE NUMBER MAKE SENSE?

Only a small share of elimination is renal, so almost all of the 0.5 L/hr is hepatic: 0.5 − 0.018 = 0.482 L/hr. Cl_H lies between Cl_R, 0.018, and Cl_T, 0.5 L/hr.

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j

Her key

Answer: 1400 (mg/L)·hr

WHAT THE STEM GIVES

Cl_T	0.5 L/hr	the denominator of AUC = $\frac{D_0}{Cl_T}$

V_D	9 L	*not needed: AUC depends on dose and clearance alone*
collection	96 hours	*not needed*
$D_u\infty$	25 mg	*not needed: the whole dose enters the AUC, not the recovered part*
D_0	700-mg	the numerator, with F = 1 for the IV dose

THE WORKING, ONE LINE AT A TIME

SETUP

$$AUC = \frac{D_0}{Cl_T}$$

$Cl_T = \frac{FD_0}{AUC}$ rearranged gives AUC = $\frac{FD_0}{Cl_T}$, and an IV dose has F = 1, so the whole 700 mg counts.

ALGEBRA

$$AUC = \frac{700\ mg}{0.5\ L/hr} = 1400\ (mg/L)\cdot hr$$

$\frac{700\ mg}{0.5\ L/hr}$ = 1400: dividing by L/hr is multiplying by hr/L, so mg becomes (mg/L)·hr, the unit of an area under a concentration–time curve. The AUC is the total exposure from the dose; at 0.5 L/hr the body clears 700 mg only after 1400 (mg/L)·hr of exposure.

> **DOES THE NUMBER MAKE SENSE?**
> Clearing 700 mg at 0.5 L/hr takes 1400 L of plasma, so the area is 1400 (mg/L)·hr; dividing by a clearance below 1 L/hr must give a number larger than the dose, 700.